Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2024

Course : B.Tech | *Time: 03 Hours*
Code : 102602 | *Subject: Dynamics of Machinery* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 *Choose the correct option / answer the following (Any seven question only):* [2 x 7 = 14]

(a) When the pitching of a ship is upward, the effect of gyroscopic couple acting on it will be
- (i) to move the ship towards port side
- (ii) to move the ship towards star-board
- (iii) to raise the bow and lower the stern
- (iv) to raise the stern and lower the bow

(b) A motor car moving at a certain speed takes a left turn in a curved path. If the engine rotates in the same direction as that of wheels, then due to the centrifugal forces
- (i) the reaction on the inner wheels increases and on the outer wheels decreases
- (ii) the reaction on the outer wheels increases and on the inner wheels decreases
- (iii) the reaction on the front wheels increases and on the rear wheels decreases
- (iv) the reaction on the rear wheels increases and on the front wheels decreases

(c) A rigid body, under the action of external forces, can be replaced by two masses placed at a fixed distance apart. The two masses form an equivalent dynamical system, if
- (i) the sum of two masses is equal to the total mass of the body
- (ii) the centre of gravity of the two masses coincides with that of the body
- (iii) the sum of mass moment of inertia of the masses about their centre of gravity is equal to the mass moment of inertia of the body
- (iv) all of the above

(d) In an engine, the work done by inertia forces in a cycle is
- (i) positive
- (ii) zero
- (iii) negative
- (iv) none of these

(e) The maximum fluctuation of energy is the
- (i) sum of maximum and minimum energies
- (ii) difference between the maximum and minimum energies
- (iii) ratio of the maximum energy and minimum energy
- (iv) ratio of the mean resisting torque to the work done per cycle

(f) In a turning moment diagram, the variations of energy above and below the mean resisting torque line is called
- (i) fluctuation of energy
- (ii) maximum fluctuation of energy
- (iii) coefficient of fluctuation of energy
- (iv) none of the above

(g) The cam follower extensively used in air-craft engines is
- (i) knife edge follower
- (ii) flat faced follower
- (iii) spherical faced follower
- (iv) roller follower

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(h) For a given lift of the follower of a cam follower mechanism, a smaller base circle diameter is desired.
(i) because it will give a steeper cam and higher pressure angle.
(ii) because it will give a profile with lower pressure angle
(iii) because it will avoid jumping
(iv) none of the above.

(i) The factor which affects the critical speed of a shaft is
(i) diameter of the disc
(ii) span of the shaft
(iii) eccentricity
(iv) all of these

(j) A shaft carrying two rotors as its ends will have
(i) No node
(ii) One node
(iii) Two nodes
(iv) Three nodes

Q.2 (a) A four-wheeled trolley car of mass 2500 kg runs on rails, which are 1.5 m apart and travels around a curve of 30 m radius at 24 km / hr. The rails are at the same level. Each wheel of the trolley is 0.75 m in diameter and each of the two axles is driven by a motor running in a direction opposite to that of the wheels at a speed of five times the speed of rotation of the wheels. The moment of inertia of each axle with gear and wheels is 18 kg-m^2. Each motor with shaft and gear pinion has a moment of inertia of 12 kg-m^2. The centre of gravity of the car is 0.9 m above the rail level. Determine the vertical force exerted by each wheel on the rails taking into consideration the centrifugal and gyroscopic effects. State the centrifugal and gyroscopic effects on the trolley. [7]

(b) The turbine rotor of a ship has a mass of 3500 kg. It has a radius of gyration of 0.45 m and a speed of 3000 r.p.m. clockwise when looking from stern. Determine the gyroscopic couple and its effect upon the ship: [7]
1) when the ship is steering to the left on a curve of 100 m radius at a speed of 36 km/h.
2) when the ship is pitching in a simple harmonic motion, the bow falling with its maximum velocity. The period of pitching is 40 seconds and the total angular displacement between the two extreme positions of pitching is 12 degrees.

Q.3 (a) A vertical petrol engine 100 mm diameter and 120 mm stroke has a connecting rod 250 mm long. The mass of the piston is 1.1 kg. The speed is 2000 r.p.m. On the expansion stroke with a crank 20° from top dead centre, the gas pressure is 700 kN/m^2. Determine: [7]
1) Net force on the piston, 2) Resultant load on the gudgeon pin,
3) Thrust on the cylinder walls, and 4) Speed above which, other things remaining same, the gudgeon pin load would be reversed in direction.

(b) A horizontal steam engine running at 240 r.p.m. has a bore of 300 mm and stroke 600 mm. The connecting rod is 1.05 m long and the mass of reciprocating parts is 60 kg. When the crank is 60° past its inner dead centre, the steam pressure on the cover side of the piston is 1.125 N/mm^2 while that on the crank side is 0.125 N/mm^2. Neglecting the area of the piston rod, determine : 1) the force in the piston rod ; and 2) the turning moment on the crankshaft. [7]

Q.4 (a) The turning moment diagram for a petrol engine is drawn to the following scales : Turning moment, 1 mm = 5 N-m ; crank angle, 1 mm = 1°. The turning moment diagram repeats itself at every half revolution of the engine and the areas above and below the mean turning moment line taken in order are 295, 685, 40, 340, 960, 270 mm^2. The rotating parts are equivalent to a mass of 36 kg at a radius of gyration of 150 mm. Determine the coefficient of fluctuation of speed when the engine runs at 1800 r.p.m. [7]

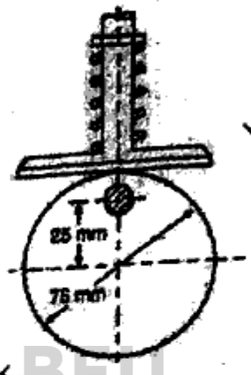
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(b) A single cylinder double acting steam engine develops 150 kW at a mean speed of 80 r.p.m. The coefficient of fluctuation of energy is 0.1 and the fluctuation of speed is ± 2% of mean speed. If the mean diameter of the flywheel rim is 2 metre and the hub and spokes provide 5% of the rotational inertia of the flywheel, find the mass and cross-sectional area of the flywheel rim. Assume the density of the flywheel material (which is cast iron) as 7200 kg/m³. [7]

Q.5 (a) A symmetrical circular cam operating a flat-faced follower has the following particulars : [7]
Minimum radius of the cam = 30 mm ; Total lift = 20 mm ; Angle of lift = 75° ; Nose radius= 5 mm ; Speed = 600 r.p.m. Find : 1. the principal dimensions of the cam, and 2. the acceleration of the follower at the beginning of the lift, at the end of contact with the circular flank , at the beginning of contact with nose and at the apex of the nose.

(b) A cam consists of a circular disc of diameter 75 mm with its centre displaced 25 mm from the camshaft axis. The follower has a flat surface (horizontal) in contact with the cam and the line of action of the follower is vertical and passes through the shaft axis as shown in Fig. [7]



The mass of the follower is 2.5 kg and is pressed downwards by a spring which has a stiffness of 3.5 N/mm. In the lowest position the spring force is 45 N.
1) Derive an expression for the acceleration of the follower in terms of the angle of rotation from the beginning of the lift.
2) As the cam shaft speed is gradually increased, a value is reached at which the follower begins to lift from the cam surface.
Determine the camshaft speed for this condition.

Q.6 (a) A shaft carries four masses A, B, C and D of magnitude 200 kg, 300 kg, 400 kg and 200 kg respectively and revolving at radii 80 mm, 70 mm, 60 mm and 80 mm in planes measured from A at 300 mm, 400 mm and 700 mm. The angles between the cranks measured anticlockwise are A to B 45°, B to C 70° and C to D 120°. The balancing masses are to be placed in planes X and Y. The distance between the planes A and X is 100 mm, between X and Y is 400 mm and between Y and D is 200 mm. If the balancing masses revolve at a radius of 100 mm, find their magnitudes and angular positions. [7]

(b) Four masses A, B, C and D are attached to a shaft and revolve in the same plane. The masses are 12 kg, 10 kg, 18 kg and 15 kg respectively and their radii of rotations are 40 mm, 50 mm, 60 mm and 30 mm. The angular position of the masses B, C and D are 60°, 135° and 270° from the mass A. Find the magnitude and position of the balancing mass at a radius of 100 mm. [7]

Q.7 (a) A cantilever shaft 50 mm diameter and 300 mm long has a disc of mass 100 kg at its free end. The Young's modulus for the shaft material is 200 GN/m². Determine the frequency of longitudinal and transverse vibrations of the shaft. [7]

(b) A shaft 50 mm diameter and 3 metres long is simply supported at the ends and carries three loads of 1000 N, 1500 N and 750 N at 1 m, 2 m and 2.5 m from the left support. The Young's modulus for shaft material is 200 GN/m². Find the [7]

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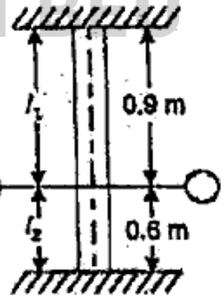
frequency of transverse vibration.

Q.8 (a) Calculate the whirling speed of a shaft 20 mm diameter and 0.6 m long carrying a mass of 1 kg at its mid-point. The density of the shaft material is 40 Mg/m^3, and Young's modulus is 200 GN/m^2. Assume the shaft to be freely supported. [7]

(b) A single-cylinder engine of total mass 200 kg is to be mounted on an elastic support which permits vibratory movement in vertical direction only. The mass of the piston is 3.5 kg and has a vertical reciprocating motion which may be assumed simple harmonic with a stroke of 150 mm. It is desired that the maximum vibratory force transmitted through the elastic support to the foundation shall be 600 N when the engine speed is 800 r.p.m. and less than this at all higher speeds. [7]
1) Find the necessary stiffness of the elastic support, and the amplitude of vibration at 800 r.p.m., and
2) If the engine speed is reduced below 800 r.p.m. at what speed will the transmitted force again becomes 600 N?

Q.9 (a) A steel shaft 1.5 m long is 95 mm in diameter for the first 0.6 m of its length, 60 mm in diameter for the next 0.5 m of the length and 50 mm in diameter for the remaining 0.4 m of its length. The shaft carries two flywheels at two ends, the first having a mass of 900 kg and 0.85 m radius of gyration located at the 95 mm diameter end and the second having a mass of 700 kg and 0.55 m radius of gyration located at the other end. Determine the location of the node and the natural frequency of free torsional vibration of the system. The modulus of rigidity of shaft material may be taken as 80 GN/m^2. [7]

(b) A flywheel is mounted on a vertical shaft as shown in Fig. [7]



The both ends of a shaft are fixed and its diameter is 50 mm. The flywheel has a mass of 500 kg and its radius of gyration is 0.5 m. Find the natural frequency of torsional vibrations, if the modulus of rigidity for the shaft material is 80 GN/m^2.

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